Cyberbullying Detection Using PCA Extracted GLOVE Features and RoBERTaNet Transformer Learning Model

Muhammad Umer, Ebtisam Abdullah Alabdulqader, Aisha Ahmed Alarfaj, Lucia Cascone, *Member, IEEE*, and Michele Nappi, *Senior Member, IEEE*

***Abstract*—Online platforms are nurturing social interactions, yet regrettably, they have also led to the proliferation of antisocial behaviors such as cyberbullying, trolling, and hate speech on a global scale. The identification of hate speech and aggression has become indispensable in the fight against cyberbullying and online harassment. Cyberbullying encompasses the use of aggressive and offensive language, including rude, insulting, hateful, and teasing comments, to inflict harm on individuals through social media platforms. Human moderation is both sluggish and costly, rendering it impractical in light of the exponential growth of data. Consequently, automated detection systems are imperative to effectively combat trolling. This study addresses the challenge of automatically discerning cyberbullying in tweets sourced from a publicly available cyberbullying dataset. The proposed methodology leverages the robustly optimized bidirectional encoder representations from transformers approach (RoBERTa), integrating principle component analysis (PCA) extracted global vectors for word representation (GLOVE) word embedding features. Furthermore, our proposed approach is benchmarked against state-of-the-art machine learning, deep learning, and transformer-based methods, utilizing the GLOVE word embedding technique. Statistical analyses reveal that our proposed model outperforms its counterparts, achieving a 0.98 accuracy and recall rate with 0.97 of precision and F1 score in detecting cyberbullying tweets. Results from *k*-fold cross validation further corroborate the superior performance of our proposed model.**

***Index Terms*—Cyberbullying detection, global vectors for word representation (GLOVE), natural language processing (NLP) for social media analysis, principle component analysis (PCA), robustly optimized bidirectional encoder representations from transformers approach (RoBERTa), transformer-based learning.**

I. Introduction

SOCIAL media platforms serve as conduits for individuals to connect and interact with their friends online, enabling the sharing of photos, videos, and daily updates. In today's digital age [1], almost everyone actively participates in social media, incorporating virtual meeting platforms into their daily routines via global online social networks (OSNs) to facilitate communication. OSNs play a pivotal role in connecting users with new friends and expanding their global connections. Moreover, data and opinion sharing are significant features of OSNs [2], [3], [4]. The utilization of OSNs has experienced exponential growth in recent years, with platforms such as Facebook, Google+, LinkedIn, Twitter, VKontakte, Mixi, and Sina Weibo emerging as preferred communication channels for billions of daily active users [5]. Statistics indicate that nearly 3.6 billion users engaged with social media networking sites in 2020, a number projected to reach 4.41 billion by 2025. According to Backlink, 58.11% of the global population uses social media [6]. Users dedicate a substantial amount of time to updating their content, interacting with primary users, and browsing others' accounts to find specific information, making social network websites an integral part of their daily activities. OSNs have the potential to eliminate economic and geographical barriers among users, fostering information sharing and communication. Additionally, OSNs play a crucial role in achieving objectives such as entertainment, education, and job searching. However, the widespread use of OSNs also increases the risk of various attacks on users. Many OSN users expose their private data, providing opportunities for attackers to engage in specific malicious activities [7], [8].

Manuscript received 20 February 2024; revised 5 May 2024; accepted 17 June 2024. Date of publication 5 September 2024; date of current version 2 October 2025. This work was supported in part by Princess Nourah bint Abdulrahman University Researchers through Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia, under Project PNURSP2024R348, in part by the project IDA included in the Spoke 2—Misinformation and Fakes of the Research and Innovation Program under Grant PE00000014, in part by "SEcurity and RIghts in the CyberSpace (SERICS)" via the National Recovery and Resilience Plan, Mission 4 "Education and Research"—Component 2 "From Research to Enterprise"—Investment 1.3 through the European Union—NextGenerationEU. *(Corresponding author: Lucia Cascone.)*

Muhammad Umer is with the Department of Computer Science & Information Technology, The Islamia University of Bahawalpur, Bahawalpur 63100, Pakistan (e-mail: umer.sabir@iub.edu.pk).

Ebtisam Abdullah Alabdulqader is with the Department of Information Technology, College of Computer and Information Sciences, King Saud University, Riyadh 11421, Saudi Arabia (e-mail: eabdulqader@ksu.edu.sa).

Aisha Ahmed Alarfaj is with the Department of Information Systems, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, Riyadh 11671, Saudi Arabia (e-mail: Aiaalarfaj@pnu.edu.sa).

Lucia Cascone and Michele Nappi are with the Department of Computer Science, University of Salerno, 84084 Fisciano, Italy (e-mail: lcascone@unisa.it; mnappi@unisa.it).

Digital Object Identifier 10.1109/TCSS.2024.3422185

© 2024 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

In tandem with the widespread use of social networks, cyberbullying has emerged as a troubling phenomenon, encompassing bullying activities that occur through digital devices such as cell phones, computers, and tablets [9], [10]. Presently, the predominant users of the internet are young people, with 95% of teenagers in the U.S. being online, and a significant majority accessing social networks through their mobile devices [11]. In the era of Web 2.0, cyberbullying is more likely to occur compared to traditional bullying [12]. A notable 36.5% of individuals feel they have experienced cyberbullying at some point in their lives, and 17.4% have reported incidents within the last 30 days [13], [14]. These figures have more than doubled since 2007 [15]. Furthermore, a 2019 Google survey revealed that teachers in the U.S. consider cyberbullying as their primary safety concern in classrooms [16], [17]. Over the past few years, online communication has increasingly become user-driven, with social network platforms emerging as the most vulnerable to cyberbullying due to their massive global user base, efficient communication channels, and continuous "24/7/365" services [18]. The impact of cyberbullying extends beyond affecting individuals' psychological well being, influencing various aspects of their lives. This is particularly concerning for young people, as cyberbullying has the potential to lead them toward self-harm and even suicide [19], [20].

Cyberbullying has emerged as a significant issue over the past decade, particularly impacting children and adolescents. Notably, a study conducted in the U.S. revealed that more than 43% of teenagers in the country have experienced cyberbullying [21]. European statistics from ER indicate that approximately 18% of youngsters in Europe have encountered bullying or harassment through the Internet and mobile phones [22]. An online report for EU kids in 2014 further highlights that every fifth student of the 11–16 years age group has faced instances of cyberbullying [23]. Even in developed countries such as Sweden, cyberbullying has increased manifold, reaching a crucial point and steadily deteriorating [24]. These findings underscore the urgency of devising effective, prompt, and proven solutions to address this pervasive issue on the Internet. Therefore, it is crucial to approach the problem of cyberbullying from various angles, including the development of automated systems for the identification and prevention of such incidents.

Given the escalating nature of this issue, researchers have actively pursued methods for the detection and prevention of cyberbullying. A particularly promising avenue involves the application of ML algorithms capable of analyzing extensive datasets to discern patterns and relationships among variables [25], [26]. Recognizing the complexity of the task at hand, it is improbable that a sole ML algorithm would prove adequate. In the context of this article, we propose a model grounded in RoBERTa to specifically identify cyberbullying in textual content. The primary contributions of this work can be succinctly summarized as follows.

1) To enhance the predictive accuracy of cyberbullying, a novel framework is proposed that is based on PCA-based feature extracted with GLOVE features and RoBERTa model. The transformer-based RoBERTa model makes use of GLOVE features to give optimal results.
2) The study involves an assessment of the performance of established machine, deep, and transformer-based learning algorithms applied to cyberbullying data. These models include RF, SVM, KNN, Naive Bayes (NB), bidirectional long short-term memory (BiLSTM), ConvLSTM, bidirectional encoder representations from transformers (BERTs), and CNN.
3) The effectiveness of the proposed approach is thoroughly examined through extensive experiments, and a comparative analysis with various state-of-the-art methods is conducted. To validate the robustness of the proposed approach, the results are further substantiated using k-fold cross validation.

The structure of this article is as follows. Section II delves into the existing research, featuring studies that utilize ML and DL for cyberbullying detection. Section III outlines the materials, methodologies, and suggested models for cyberbullying identification. Section IV provides comprehensive details about the experiment's results and their corresponding discussions. Finally, Section V concludes the article with summarizing remarks.

II. Literature Review

The phenomenon of cyberbullying on social media platforms, particularly prominent on platforms such as Twitter (referred to as X) and Facebook, is a matter of significant concern due to its profound impact on users' well being, especially among younger demographics who are frequent users of these platforms. Ottosson [27] introduced an LLM aimed at detecting cyberbullying on social media platforms, utilizing the GPT-3 LLM to bridge the gap in platform moderation. The study demonstrates that fine-tuning an LLM proves to be an effective strategy for enhancing cyberbullying detection, achieving an accuracy of 90%. In another study by Alhloul and Alam [28], a DL-based system was proposed for identifying bullying tweets. The researchers developed a CNN-attention framework that combines an attention layer with a convolutional pooling layer to efficiently extract cyberbullying-related keywords from users' tweets, outperforming other learning models with an accuracy of 97.10%.

Wang et al. [29] presented a graphical convolutional method for detecting cyberbullying, leveraging a semisupervised online dynamic query expansion (DQE) process to generate balanced data. They introduced a graph convolutional network (GCN) classifier that operates on a graph constructed from thresholded cosine similarities between tweet embeddings, achieving an accuracy score of 92.70% with the proposed SOSNet using SBERT. In a separate study, Qudah et al. [30] suggested an improved system for cyberbullying detection utilizing an adaptive external dictionary (AED), demonstrating superior accuracy in detecting cyberbullying incidents.

Mathur et al. [31] introduced a real-time system for detecting cyberbullying on Twitter using natural language processing (NLP) and ML, achieving an impressive accuracy of 94.06% with the RF algorithm through careful preprocessing and tuning. Bokolo and Liu [32] proposed a DL-based

system for automatically detecting cyberbullying on social media platforms, with the Bi-LSTM model outperforming others with a remarkable accuracy of 98%. Nisha and Jebathangam [33] introduced an automated system for classifying and detecting cyberbullying on social media, achieving an accuracy score of 99.1% with the proposed DT classifier. Mehmud et al. [34] proposed an ML-based system focusing on textual features for cyberbullying detection, with LGBM demonstrating high accuracy of 85.5%. Muneer et al. [35] proposed a modified BERT and stacking ensemble model achieving an outstanding accuracy of 97.4%, while another study focusing on attention-based Conv1DLSTM classifier attained the highest accuracy of 94.49%.

The proposed research work is an extension of the work [36] done by common coauthor Muhammad Umer in improving the accuracy of the proposed framework. This extension not only enhances the outcomes but also diminishes the computational complexity of the proposed model. It incorporates a LIME module for explainability, which makes the model's predictions more accessible and comprehensible. Additionally, it includes a statistical analysis that validates the model's superior performance over traditional classifiers, thereby underscoring the effectiveness and efficiency of the approach adopted. Adding PCA as a preprocessing step with RoBERTaNet model for text classification enhances the performance of GLOVE embeddings for several reasons. First, PCA reduces the dimensionality of GLOVE embeddings by 64%, which helps alleviate the curse of dimensionality. By reducing the number of features while preserving the most relevant information, PCA enables the model to focus on the most important aspects of the data, leading to improved performance. Additionally, PCA helps to mitigate the issue of multicollinearity among the word embeddings. Word embeddings often exhibit high correlations with each other due to the inherent structure of language. This multicollinearity negatively impacts the performance of ML models, leading to instability and overfitting. PCA addresses this issue by transforming the original embeddings into a set of orthogonal components, thereby reducing redundancy and improving the model's robustness. Moreover, PCA enhances the interpretability of the word embeddings by capturing the underlying structure of the data in a more concise and meaningful representation. By retaining the principal components that explain the most variance in the data, PCA enables the model to focus on the most salient aspects of the word embeddings, making them more informative and discriminative for the text classification task. Overall, adding PCA as a preprocessing step with RoBERTaNet model for text classification can improve the performance of GLOVE embeddings by reducing dimensionality, mitigating multicollinearity, and enhancing interpretability. This can lead to more accurate and reliable text classification models that effectively leverage the semantic information encoded in the word embeddings.

A summary of state-of-the-art related works is presented in Table I. In the literature summary of Table I, we can observe that recent research in this field primarily utilizes single-feature engineering techniques in conjunction with individual ML or DL models. While some studies have explored combinations of layers, such as CNN-LSTM, it is important to note that such combinations are typically beneficial for feature extraction rather than independent voting mechanisms. In our research, we employ diverse feature engineering techniques GLOVE, FastText, and word2vec to identify the most effective features and then use the RoBERTaNET model to accurately classify the cyberbullying tweets. This comprehensive approach allows us to achieve superior results compared to the existing state-of-the-art research in the field.

TABLE I
SUMMARY OF STATE-OF-THE-ART RELATED WORK

Reference	Classifiers	Performance
[27]	BERT, GPT-3	90% GPT-3 Ada
[28]	LR, RF, XGB, CNN, CNN+LR, CNN+RF, CNN+XGB, CNN+LSTM, CNN-Attention	97.10% CNN-Attention
[29]	LR, NB, KNN, SVM, XGB, MLP, SOSNet	92.70% SOSNet with SBERT
[30]	RF, XGB, CatBoost, Ensemble Voting	98.55% Ensemble voting
[31]	Vanilla RF, AdaBoost, GBC, Tuned RF	94.06% Tuned RF
[32]	NB, SVM, BI-LSTM	98% Bi-LSTM
[33]	SVM, ANN, DT	99.1% DT
[34]	LGBM, XGB, LR, RF, ADA	85.5% LGBM
[35]	Conv1DLSTM, BiLSTM, LSTM, BERT, Tuned-BERT, Stacked and CNN	97.4% stacked
[37]	LSTM, Conv1DLSTM, CNN, BiLSTM, BiLSTM_Pooling, GRU	94.49% Conv1DLSTM
This work	RoBERTa with GLOVE and PCA features	98.7%

III. MATERIAL AND METHODS

This section delineates the essential components of the proposed framework for cyberbullying detection, named cyberbullying detection system (CDS). It delves into the analysis and identification of cyberbullying instances across various social media platforms. Furthermore, this section elaborates on the dataset utilized, ML and DL models employed in the study, and provides a succinct overview of the feature extraction techniques incorporated.

A. Dataset

As social media usage becomes ubiquitous across diverse age groups, a significant portion of the population relies on this essential tool for daily communication. The pervasive nature of social media has led to cyberbullying becoming a widespread issue, capable of impacting individuals regardless of time and place. The inherent anonymity of the internet adds to the complexity of addressing such attacks compared to traditional forms of bullying. The recent COVID-19 pandemic, characterized by widespread school closures, reduced in-person social interactions, and increased screen time, has heightened concerns about exposure to cyberbullying. Statistics indicate that 36.5% of middle and high school students have experienced cyberbullying,

TABLE II
DATASET EXAMPLES

Text	Label
Thanks for the heads up, but not too concerned about another angry dude on Twitter	Not Cyberbullying
The girls are gonna be pissed about this matter	Cyberbullying

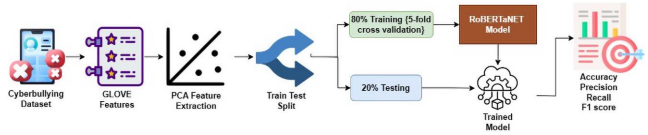

Fig. 1. Architecture of proposed RoBERTaNet model.

significantly affecting academic performance, inducing feelings of sadness, and even leading to suicidal ideation.

The dataset under consideration is sourced from the Kaggle website and comprises over 47 000 tweets classified as cyberbullying[1] [38]. The categories within this dataset include: gender, age, religion, ethnicity, other forms of cyberbullying, and no online bullying. It is noteworthy that the dataset is balanced, with around 7850 instances in each class, ensuring equal representation. The average length of each tweet is 13 words. Upon exploration, it is observed that the tweets are either entirely offensive or describe incidents of bullying. Some of the dataset examples are shown in Table II.

B. PCA

PCA, renowned for its capability to reduce the dimensionality of attribute sets through linear transformation, finds utility across various domains. This technique streamlines the resulting dataset for enhanced readability while retaining the crucial features of the original data [39]. The dimensionality of the new dataset may vary from that of the original, either increasing or decreasing. PCA relies on the covariance matrix during computation, prioritizing components from most significant to least significant [40]. The transformation of the original matrix, presumed to have a dimensions and b observations, into a t-dimensional subspace is described by the following equation:

$$Y = (E^T X). \tag{1}$$

Here, $E_{a\times t}$ represents the projection matrix comprising t eigenvectors corresponding to the t largest eigenvalues. $X_{a\times b}$ denotes the mean-centered data matrix. The number of features generated with GLOVE word embedding is 25 000 while after PCA significant feature extraction the number of features is set to 9000.

C. Feature Extraction Methods

Feature extraction methods were applied separately to the training and testing datasets. These techniques were used to train the selected models on the training data and subsequently utilized during the classification of the testing data. In the context of cyberbullying detection in this study, three feature extraction methods, namely GLOVE, Word2Vec, and FastText, were utilized. A brief overview of these techniques is provided as follows.

1) Word2Vec: To enhance word embedding, Word2Vec, a robust statistical model, was developed. Word representations were elevated through Word2Vec, allowing the model to capture subtle distinctions efficiently [41]. These representations significantly influence semantics, depicting relationships between words. Word2Vec includes two models: the CBoW and the continuous skip-gram model.

2) GLOVE: GLOVE was introduced to create a more efficient model that integrates the strengths of various approaches. It operates on the entire corpus by employing a word context matrix and a word cooccurrence matrix.

3) FastText: FastText is a word embedding library that encompasses 300-dimensional vectors for 2 million common crawl words [42]. It leverages morphological features to identify challenging words, enhancing its suitability for vector representation.

D. Proposed Methodology

This section introduces the framework proposed for this study, as depicted in Fig. 1. A growing trend in using DL-based classifiers has emerged, with the RoBERTa model gaining attention for its potential to enhance traditional classifiers for classification accuracy. Therefore, this study aims to leverage RoBERTa to detect cyberbullying in text data.

Experiments are conducted using benchmark datasets related to cyberbullying, employing various word embedding techniques, including Word2Vec and FastText. Subsequently, the proposed method, which combines PCA extracted GLOVE features with RoBERTa, is applied for training. The effectiveness of the proposed methodology is evaluated using precision, recall, accuracy, and F1 score.

1) RoBERTa: RoBERTaNet, an extension of the BERT model, incorporates several enhancements to improve its effectiveness in natural language understanding tasks [43]. These enhancements include larger batch sizes, dynamic masking strategies, and longer training times, which enable RoBERTaNet to achieve state-of-the-art performance on various NLP benchmarks. By examining the intricate details of RoBERTaNet's architecture, such as its multilayer transformer architecture, attention mechanisms, and pretraining objectives, researchers can gain valuable insights into the model's underlying mechanisms and better understand its capabilities.

A comprehensive analysis of RoBERTaNet's functionality can shed light on the specific features and techniques that contribute to its success. For example, researchers can explore how RoBERTaNet's self-attention mechanism enables it to capture long-range dependencies and contextual information in text data. Additionally, insights into the pretraining strategies and data augmentation techniques used in training RoBERTaNet can provide valuable guidance for improving the performance of similar models. By conducting a detailed examination of RoBERTaNet's architecture and functionality, researchers can

[1] https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification

uncover the key factors driving its superior performance and pave the way for further advancements in NLP. RoBERTa demonstrates superior performance across a spectrum of NLP tasks, including question answering, sentiment analysis, and named entity recognition, due to its robustness and versatility.

E. Selected Baseline Methods

This section discusses the ML algorithms utilized for cyberbullying detection using the Twitter dataset. The implementation of these ML models involves the Sci-Kit Learn library and NLTK. Specifically, four ML algorithms and three DL algorithms were applied in Python. To evaluate the effectiveness of the proposed system, a combination of regression-based, tree-based, and ensemble-based models was utilized. The following ML algorithms, integrated with the proposed methodology, were employed to assess the performance of the ML classifiers.

1) RF: RF represents an ensemble of decision trees (DTs). It operates as a supervised learning algorithm, employing multiple DTs that work independently to predict a class label. The final prediction is determined by the class receiving the majority of votes from the individual trees [44]. RF exhibits notably low error rates when compared to other models, attributed to the reduced correlation between the constituent trees. In this study, RF was trained using various parameters, and for DT splits, multiple algorithms were employed depending on the specific problem at hand.

2) KNN: KNN is categorized as a lazy learner as it requires all the data during the training process. In KNN, a new sample is assigned to a class using a distance metric [45]. KNN calculates the distance of new data points to their closest neighbors, where the value of K dictates the number of neighbors considered. For instance, if K is set to 3, new data points are assigned to a class based on their distance to the three nearest neighbors. The distance calculation between two points can be performed using various distance estimation metrics, with the most commonly used ones including Euclidean, Manhattan, and Minkowski distances.

3) NB: NB, a variant grounded in Bayes' theorem, operates on conditional probabilities to predict the outcome of an event [46]. In this context, if a sample is categorized into k classes represented by $k = \{c1, c2, \ldots, ck\}$, the output results are denoted as c. The NB function is outlined as follows, where c represents the class and d signifies the sample

$$P(c|d) = \frac{P(d|c) \cdot P(c)}{P(d)}. \tag{2}$$

This formula calculates the probability of class c given sample d, which is the product of the likelihood of the sample given the class $[P(d|c)]$ and the prior probability of the class $[P(c)]$, normalized by the probability of the sample $[P(d)]$.

4) SVM: SVM aims to adapt to the provided data and determine the optimal hyperplane that effectively separates the data [47]. Once this hyperplane is obtained, certain features are input into the classifier to derive the final class results. This unique characteristic makes SVM a distinctive algorithm. SVM employs a linear kernel function and is implemented in liblinear rather than libSVM. Notably flexible in terms of penalty and loss function choices, SVM yields favorable scaling results, particularly for large sample sizes. As a widely utilized supervised learning algorithm, SVM excels in handling both classification and regression problems. The SVM algorithm can be implemented with various kernel functions to suit different types of data distributions. Within the SVM framework, there are options for employing a linear kernel, known as linear SVM, as well as other kernels for more complex data separations, collectively referred to as SVM with nonlinear kernels.

5) BiLSTM: BiLSTM is a subtype of recurrent neural network (RNN) specifically designed for handling sequence processing tasks, commonly applied in NLP and speech recognition [48]. Unlike traditional LSTMs, BiLSTM processes input data in both forward and backward directions, allowing it to capture contextual information from both preceding and subsequent states. This bidirectional strategy enhances the network's ability to comprehend and learn long-range dependencies within sequential data. Comprising two LSTM layers, one processing the input sequence from start to finish and the other in reverse, BiLSTM concatenates the outputs from both directions. This amalgamation yields a more comprehensive representation of the input sequence, making BiLSTM particularly adept at tasks requiring a nuanced understanding of context and dependencies in both directions.

6) ConvLSTM: ConvLSTM represents a neural network architecture that merges the spatial hierarchies learned by convolutional layers with the ability to capture temporal dependencies provided by LSTM networks [49]. This architecture is particularly effective in handling spatiotemporal data, such as video sequences or time-series data characterized by spatial dependencies. ConvLSTM seamlessly incorporates convolutional operations into the LSTM cells, enabling the model to simultaneously learn both spatial patterns and temporal dynamics. This integration makes ConvLSTM well suited for tasks demanding an understanding of intricate patterns in both spatial and temporal dimensions, including applications such as video prediction, anomaly detection, and weather forecasting.

7) CNN: CNN is a prevalent artificial neural network employed across a diverse range of tasks [50], [51]. Conceptually, CNN resembles a multilayer perceptron (MLP) with an activation function for each neuron, mapping the weighted outputs. Unlike MLPs, CNNs are invariant to translation and rotation due to their architecture. CNNs consist of convolutional layers, pooling layers, and fully connected layers, each incorporating an activation function.

8) BERT: BERT is a transformer-based model featuring an attention mechanism that actively processes input text [52]. Comprising an encoder and a decoder, BERT takes textual input and generates predictions as output. BERT excels in tasks such as question answering and sentiment analysis, owing to its training on extensive word-based data. Unlike traditional models, BERT analyzes the context of words in both directions simultaneously through its attention mechanism, offering a nuanced understanding of word meanings.

TABLE III
MODELS RESULTS USING WORD2VEC FEATURES

Classifier	Accuracy	Precision	Recall	F1 Score
RF	0.92	0.91	0.89	0.90
KNN	0.80	0.76	0.70	0.73
NB	0.85	0.82	0.83	0.82
SVM	0.84	0.83	0.81	0.82
BiLSTM	0.85	0.82	0.83	0.82
Conv LSTM	0.87	0.83	0.84	0.84
BERT	0.91	0.90	0.91	0.91
CNN	0.82	0.79	0.76	0.77
RoBERTa	**0.93**	**0.92**	**0.93**	**0.92**

Note: The highest values are highlighted in bold.

TABLE IV
MODELS RESULTS USING GLOVE FEATURES

Classifier	Accuracy	Precision	Recall	F1 Score
RF	0.96	0.95	0.97	0.96
KNN	0.87	0.89	0.87	0.88
NB	0.88	0.88	0.89	0.88
SVM	0.86	0.89	0.90	0.89
BiLSTM	0.88	0.86	0.88	0.87
Conv LSTM	0.89	0.88	0.87	0.87
BERT	0.95	0.95	0.95	0.95
CNN	0.86	0.84	0.83	0.83
RoBERTa	**0.98**	**0.97**	**0.98**	**0.97**

Note: The highest values are highlighted in bold.

IV. RESULTS AND DISCUSSION

This section offers comprehensive insights into the experiments conducted using the Twitter dataset for bullying detection employing DL, ML, and transformer-based models. The evaluation of research experiments is carried out using a Python-based Colab Notebook, utilizing Tensorflow, Keras, and Scikit-learn libraries. To assess the performance of various models, multiple metrics, including precision, recall, accuracy, and the F1 score, are utilized. The experiments are being conducted on a Jupyter Notebook running on a Dell PowerEdge T430 GPU with 32 GB of memory, operating on the Windows 10 platform. This system is powered by two Intel Xeon processors, each comprising eight cores running at a clock speed of 2.4 GHz, and is supported by 32 GB of DDR4 RAM. Python 3.8 is utilized to implement all learning models. In particular, transformer-based and deep models are being trained using a GPU and 16 GB of RAM to enhance computational efficiency. The subsequent sections detail the outcomes of these experiments, where the performance of the models is evaluated using metrics such as the F1 score, accuracy, precision, and recall.

A. Model Results Using Word2Vec Features

In this study, a series of experiments are conducted to detect cyberbullying tweets, employing a combination of ML and DL models. The identification of bullying instances in the data was facilitated by incorporating PCA extracted Word2vec features into the feature set. The results of employing the Word2vec technique for multiclass classification are presented in Table III.

Table III illustrates the results obtained from utilizing Word2Vec features. Notably, RoBERTa achieves the highest accuracy of 93%, along with top values for precision, recall, and F1 scores. In the domain of ML-based classification, the tree-based model RF achieves an accuracy of 92%, accompanied by corresponding values of 91% for precision, 89% for recall, and 90% for F1 score. Among the DL models, the CNN for multiclass grouping records the lowest accuracy at 82%. Conversely, the ML model KNN emerges as the least-performing classifier, achieving an accuracy of 80%.

B. Results Using GLOVE Features

Numerous experiments were conducted to identify cyberbullying tweets, employing both ML and DL models in this study.

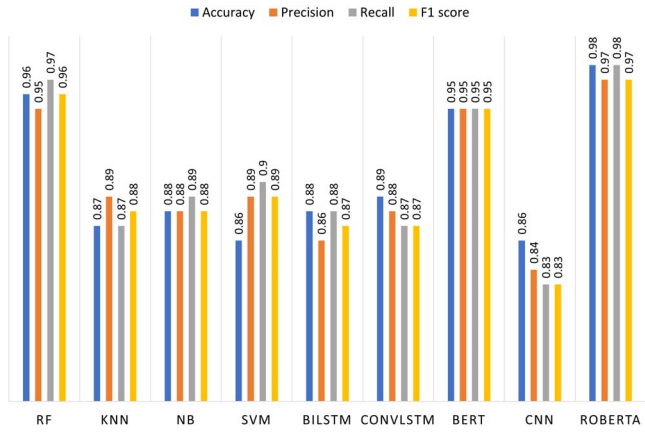

Fig. 2. Classification report of all learning models using GLOVE features.

The detection of bullying in the data was facilitated by utilizing GLOVE features as part of the feature set. The outcomes of the GLOVE technique for multiclass classification are presented in Table IV and in Fig. 2.

Table IV presents the outcomes obtained using PCA extracted GLOVE features. Notably, RoBERTa achieves the highest accuracy of 98%, and similarly, it secures top values for precision, recall, and F1 scores. In the realm of ML-based classification, the tree-based model RF attains an accuracy score of 96%, accompanied by corresponding values of 95% for precision, 97% for recall, and 96% for F1 score. Among the DL models, CNN for multiclass grouping records the lowest accuracy, standing at 86%. Conversely, the ML model SVM emerges as the least-performing classifier, achieving an accuracy of 86%.

C. Results Using FastText Features

In the subsequent set of experiments, trials were conducted utilizing PCA extracted FastText features. The outcomes for both ML and DL models, incorporating FastText features, are showcased in Table V. The results suggest a decline in the performance of the learning models. Additionally, the findings underscore that models employing Word2vec features outperform those utilizing FastText features in terms of performance.

Table V illustrates the results obtained using FastText features. RoBERTa stands out with the highest accuracy of 92.25%, along with superior values for precision, recall, and F1 scores. In the context of multiclass classification, the tree-based

TABLE V
MODELS RESULTS USING FASTTEXT FUNCTIONS

Classifier	Accuracy	Precision	Recall	F1 Score
RF	0.90	0.85	0.88	0.87
KNN	0.76	0.75	0.72	0.74
NB	0.78	0.75	0.73	0.74
SVM	0.80	0.81	0.84	0.83
BiLSTM	0.77	0.74	0.79	0.77
Conv LSTM	0.86	0.88	0.90	0.89
BERT	0.84	0.87	0.86	0.86
CNN	0.79	0.76	0.74	0.75
RoBERTa	**0.95**	**0.91**	**0.93**	**0.92**

Note: The highest values are highlighted in bold.

model RF achieves an accuracy of 88.97%, with corresponding values of 84% for precision, 87% for recall, and 85% for the F1 score. Among the DL models, CNN for multiclass classification records the lowest accuracy on FastText features, standing at 74%. Conversely, the ML model KNN again emerges as the least-performing classifier, achieving an accuracy of 74%. Notably, KNN exhibits improved performance with FastText features compared to its performance with Word2Vec features. In summary, the overall performance of the learning models demonstrates a decline using FastText features when compared to Word2vec features. Let us understand the 2% error rate with a misclassified example. Consider this tweet extracted from the dataset *Education Nation: Bullying—Turn to 10*. In this tweet, Education Nation is a TV program in which there is a funny segment called Bullying. This Bullying segment has nothing to do with bullying people but just a segment to discuss something bullish happening in the world. In this tweet, this segment is turning ten-years old and there is a tweet to celebrate this; but due to the addition of the keyword Bullying, the model considered this tweet in the cyberbullying label. This is done because the model was trained on examples where most tweets contain words such as Bullying to disrespect someone.

D. Statistical Analysis

We have identified our top-performing models, RoBERTaNet and RF, both utilizing GLOVE features. To discern any disparities in their effectiveness, we have opted to undertake a statistical analysis. Our aim is to gauge whether there is a substantial variance between their performance metrics, including accuracy, precision, recall, and F score.

For this analysis, we will employ a tenfold cross-validated paired t-test, a widely accepted approach for model comparison [53]. In this procedure, the models undergo assessment on different datasets under the same random seed, ensuring consistency in evaluation.

Let us establish our null hypothesis (H_0) and alternative hypothesis (H_1) as follows.

1) H_0: There is no notable difference in performance metrics between RoBERTaNet and RF.
2) H_1: A significant difference exists in performance metrics between RoBERTaNet and RF.

During each iteration of k-fold cross validation, we will compute the performance difference between RoBERTaNet and RF, generating k difference measures. Assuming these differences are independently drawn and conform to an approximately normal distribution, we will calculate the t statistic with $k-1$ degrees of freedom, following t-test. Our chosen significance level (α) is 0.05.

TABLE VI
K-FOLD CROSS-VALIDATION RESULTS

K-Fold	Accuracy	Precision	Recall	F1 score
Fold 1	0.94	0.95	0.97	0.96
Fold 2	0.94	0.95	0.97	0.96
Fold 3	0.95	0.96	0.97	0.96
Fold 4	0.95	0.97	0.98	0.98
Fold 5	0.94	0.96	0.98	0.97
Average	**0.95**	**0.96**	**0.97**	**0.97**

Note: The highest values are highlighted in bold.

Let $p^{(i)}$ compute the difference between the model performances in the ith iteration, $p^{(i)} = p^{(i)}_{\text{RoBERTaNet}} - p^{(i)}_{\text{RF}}$, and let $\bar{p}$ represents the average difference between the classifier performances

$$\bar{p} = \frac{1}{k}\sum_{i=1}^{k} p^{(i)}. \tag{7}$$

The t-statistic is given by

$$t = \frac{\bar{p}\sqrt{k}}{\sqrt{\sum_{i=1}^{k}(p^{(i)} - \bar{p})^2/(k-1)}}. \tag{8}$$

Once we have computed the t-statistic, we can compute the p-value and compare it to our chosen significance level. If the p-value is smaller than α, we reject the null hypothesis and accept that there is a significant difference between the two models. We will repeat these calculations for precision, recall, and F score. In our case, there is a difference between RoBERTaNet and RF results because in each case our value is smaller than 0.05.

E. K-Fold Cross-Validation Results

K-fold cross validation was utilized to enrich the investigation of the models. Fivefold cross-validation results in Table VI showcase the superior performance in terms of F1 score, accuracy, recall, and precision of the model compared to other alternatives.

F. LIMEs

LIME has established itself as a powerful asset within the realm of XAI, with a specific emphasis on text classification endeavors. Its operational framework revolves around the generation of perturbed instances of input data, followed by a meticulous observation of the resultant alterations in the model's predictions. Tailored specifically for text classification tasks, LIME offers localized, human-interpretable explanations for individual predictions, thereby empowering users to gain insights into the underlying rationale behind specific decisions. Particularly within the domain of NLP, LIME excels in identifying pivotal terms or phrases within a document that

exert a significant influence on the classification outcome. This interpretability plays a pivotal role in cultivating trust in AI models, particularly in sectors where transparency and accountability are of paramount importance, such as healthcare and finance [54].

Moreover, LIME's adeptness in elucidating the decision-making processes of complex models enhances its versatility across a wide array of applications and underscores the importance of responsible AI deployment practices. Its profound contributions to transparency and interpretability render it indispensable for researchers, practitioners, and stakeholders alike, all of whom seek to gain a comprehensive understanding of and validate the outcomes generated by text classification models [55]. In the LIME explanation, each word weight has been listed with a color coding scheme to present which feature contributes to the prediction of which class. In our case, orange color-coded words with positive weights contribute toward *Not Cyberbullying* class while blue color-coded represents the word contributes toward *Cyberbullying* class with negative weights.

1) Example: Consider the following English Cyberbulling Tweet classified by our model.

```
Best pickup line? Hi, you're cute?:
I love how people call James Potter
is a bully.
```

We used the `LimeTextExplainer` from the LIME library to generate explanations for this specific instance. The Python code snippet used for generating the LIME explanation is given as follows.

```
from lime.lime_text import LimeTextExplainer
...
# Initialize the explainer with class names for
    clarity
explainer = LimeTextExplainer(class_names=Target
    Class)
```

The model's output along with the LIME explanation is as follows.

1) Document ID: 0;
2) Probability (Not Cyberbullying) = 0.915; and
3) True class: Not Cyberbullying

The classifier accurately predicted the instance as "Not Cyberbullying." The explanation of the weighted features that influenced this prediction is given as follows.

```
[ 'Best', 0.006432,
 'pickup', 0.074982,
 'line', 0.057682,
 'Hi', 0.048863,
 'you're', 0.064129,
 'cute', 0.046281,
 'I', 0.054169,
 'love', 0.085964,
 'how', 0.022651,
 'people', 0.005419,
 'call', 0.005149,
 'James', 0.006581,
 'Potter', 0.004169,
 'is', 0.025940,
 'a', 0.051648,
 'bully', -0.075290]
```

Each word's contribution to the model's prediction is color-coded: orange for a positive contribution toward the "Not Cyberbullying" class, and blue for a negative contribution, indicating the "Cyberbullying" class.

G. Challenges of Proposed Model

Cyberbullying text classification using GLOVE's word embedding and the RoBERTaNET model faces several challenges and limitations.

1) Data Sparsity: One major challenge is the scarcity of labeled cyberbullying data, which makes it difficult to train accurate models. GLOVE's word embeddings and RoBERTaNET require large amounts of labeled data to learn meaningful representations, and the limited availability of such data can hinder the model's performance.
2) Model Complexity and Resource Intensiveness: Both GLOVE's word embedding and RoBERTaNET are complex models that require substantial computational resources for training and inference. This complexity can make the training process time-consuming and resource-intensive, especially when dealing with large datasets. For this reason, in this work, before using them as input for RoBERTaNET, the GLOVE embeddings were subjected to a process of dimensional reduction by PCA.
3) Fine-tuning Difficulty: Fine-tuning RoBERTaNET for cyberbullying text classification may pose challenges due to the need for careful parameter tuning and optimization. Finding the right hyperparameters and training settings to achieve optimal performance can be a daunting task, especially given the lack of standardized guidelines for fine-tuning.
4) Interpretability: Despite their effectiveness, models such as RoBERTaNET are often considered black-box models, making it challenging to interpret their decisions. Understanding how the model arrives at its predictions, especially in the context of cyberbullying text classification, is crucial for building trust and ensuring transparency. To address this issue, an initial attempt was made by integrating LIME as a complement. This allows users to gain a deeper understanding of the reasons behind the model's predictions.
5) Domain Specificity: GLOVE's word embedding and RoBERTaNET may not capture the nuances and context-specific features of cyberbullying text effectively. These models are trained on large corpora of general text data and may struggle to generalize well to the domain of cyberbullying, where the language and patterns can be highly specific and context-dependent.

H. Comparison of Performance With Previous Studies

The effectiveness of the proposed method was assessed by comparing its results with various approaches reported in the literature for cyberbullying detection. In this study, a range of both DL and ML models were employed to achieve meaningful results. For instance, in [32], bidirectional LSTM models were

TABLE VII
COMPARISON OF PERFORMANCE OF THE PROPOSED APPROACH WITH STATE-OF-THE-ART TECHNIQUES

Reference	Proposed Approach	Performance
[27]	GPT-3 Ada	90.2%
[29]	SOSNet with SBERT	92.7%
[31]	Tuned RF	94.6%
[34]	LightGBM	85.5%
[56]	BERT	97.34%
[56]	LSTM	92%
Proposed	**RoBERTa with GLOVE and PCA features**	**98.7%**

Note: The highest values are highlighted in bold.

utilized to predict cyberbullying incidents on social media platforms. Furthermore, Wang et al. [29] adopted a feature selection technique to enhance the performance of their models. Other studies, including [31], [33], and [34], focused on employing ML models for similar objectives. Ensemble models were also investigated in [28] and [30], contributing significantly to the effective detection of cyberbullying. The comparative results presented in Table VII indicate that the proposed approach outperforms these existing methods.

V. CONCLUSION

This research investigated the realm of cyberbullying detection in tweets, introduced an innovative method that merged the robust RoBERTa model with PCA extracted GLOVE features. By integrating a LIME module, the model's predictions became more transparent and comprehensible. Through thorough evaluation, we compared our model's performance with various existing techniques, including traditional ML, DL, and transformer-based architectures. Our extensive experimentation demonstrated the effectiveness of our model, significantly improving cyberbullying detection accuracy compared to established benchmarks. Furthermore, statistical analysis confirmed the superiority of our model over traditional classifiers, validating our approach's efficacy. Its capacity to harness contextual embeddings and nuanced linguistic attributes provided by RoBERTa, along with the complementary information encoded in GLOVE features, contributed to a robust and efficient detection mechanism. This study not only pushed forward the frontier of cyberbullying detection but also emphasized the significance of employing state-of-the-art models to tackle contemporary challenges in online social environments. Our analyses showcased our model's superior performance, achieving an accuracy and recall rate of 0.98, along with a precision of 0.97 and an F1 score, in detecting cyberbullying tweets. Results from k-fold cross validation further affirm the reliability of our approach. As online dynamics continue to evolve, our research paved the way for refining models capable of adapting to the changing landscape of cyberbullying across diverse platforms and contexts. Future endeavors may focus on tailoring DL models for small datasets and incorporating multiple datasets to construct a comprehensive and multidimensional dataset for further experimentation with our proposed approach.

REFERENCES

[1] V. Bharti, A. Kumar, V. Purohit, R. Singh, A. K. Singh, and S. K. Singh, "A label-efficient semi self-supervised learning framework for IoT devices in industrial process," *IEEE Trans. Ind. Inform.*, vol. 20, no. 2, pp. 2253–2262, Feb. 2024.

[2] S. R. Sahoo and B. B. Gupta, "Classification of various attacks and their defence mechanism in online social networks: A survey," *Enterprise Inf. Syst.*, vol. 13, no. 6, pp. 832–864, 2019.

[3] R. Annamalai, S. J. Rayen, and J. Arunajsmine, "Social media networks owing to disruptions for effective learning," *Procedia Comput. Sci.*, vol. 172, pp. 145–151, 2020.

[4] M. Fire, G. Katz, and Y. Elovici, "Strangers intrusion detection detecting spammers and fake profiles in social networks based on topology anomalies," *Human J.*, vol. 1, no. 1, pp. 26–39, 2012.

[5] A. K. Singh, B. Kumar, S. K. Singh, S. Ghrera, and A. Mohan, "Multiple watermarking technique for securing online social network contents using back propagation neural network," *Future Gener. Comput. Syst.*, vol. 86, pp. 926–939, 2018.

[6] B. Dean, "How many people use social media in 2021? (65+ statistics)." Backlinko. Accessed: Jan. 5, 2024. [Online]. Available: https://backlinko.com/social-media-users

[7] D. Paulraj, "A gradient boosted decision tree based sentiment classification of Twitter data," *Int. J. Wavelets Multiresolution Inf. Process.*, vol. 18, no. 4, pp. 2050027–2050121, 2020.

[8] H. Kefi and C. Perez, "Dark side of online social networks: Technical, managerial, and behavioral perspectives," *Encyclopedia Social Netw. Anal. Mining*, vol. 143, pp. 535–556, 2018.

[9] "What is cyberbullying." Stopbullying. Accessed: Jan. 5, 2024. [Online]. Available: https://www.stopbullying.gov/cyberbullying/what-is-it

[10] A. M. Abbas, "Social network analysis using deep learning: Applications and schemes," *Social Netw. Anal. Mining*, vol. 11, no. 1, 2021, Art. no. 106, doi: 10.1007/s13278-021-00799-z.

[11] S. Hinduja and J. W. Patchin, "Cyberbullying: Identification, prevention, and response," Cyberbullying Research Center. Accessed: Jan. 5, 2024. [Online]. Available: https://cyberbullying.org/Cyberbullying-Identification-Prevention-Response.pdf

[12] G. Kumar, J. P. Singh, and A. K. Singh, "Autoencoder-based feature extraction for identifying hate speech spreaders in social media," *IEEE Trans. Comput. Social Syst.*, early access, doi: 10.1109/TCSS.2023.3240098.

[13] "2019 cyberbullying data." Cyberbullying Research Center. [Online]. Available: https://cyberbullying.org/2019-cyberbullying-data

[14] N. Badri, F. Kboubi, and A. H. Chaïbi, "Combining FastText and glove word embedding for offensive and hate speech text detection," in *Proc. Knowl.-Based Intell. Inf. & Eng. Syst.: Proc. 26th Int. Conf. (KES)*, Verona, Italy and Virtual Event, in Procedia Computer Science, vol. 207, Amsterdam, The Netherlands: Elsevier, 2022, pp. 769–778, doi: 10.1016/j.procs.2022.09.132.

[15] "Summary of our cyberbullying research (2007–2019)." Cyberbullying Research Center. Accessed: Jan. 5, 2024. [Online]. Available: https://cyberbullying.org/summary-of-our-cyberbullying-research

[16] "Be internet awesome: Online safety & parents." Google. [Online]. Available: https://beinternetawesome.withgoogle.com/en_us

[17] D. Dessì, M. Dragoni, G. Fenu, M. Marras, and D. R. Recupero, "Evaluating neural word embeddings created from online course reviews for sentiment analysis," in *Proc. 34th ACM/SIGAPP Symp. Appl. Comput. (SAC)*, Limassol, Cyprus, C. Hung and G. A. Papadopoulos, Eds., New York, NY, USA: ACM, 2019, pp. 2124–2127, doi: 10.1145/3297280.3297620.

[18] M. A. Al-Garadi et al., "Predicting cyberbullying on social media in the big data era using machine learning algorithms: Review of literature and open challenges," *IEEE Access*, vol. 7, pp. 70701–70718, 2019.

[19] A. John et al., "Self-harm, suicidal behaviours, and cyberbullying in children and young people: Systematic review," *J. Med. Internet Res.*, vol. 20, 2018, Art. no. e129.

[20] C. Wang, P. Nulty, and D. Lillis, "A comparative study on word embeddings in deep learning for text classification," in *Proc. 4th Int. Conf. Natural Lang. Process. Inf. Retrieval (NLPIR)*, Seoul, Republic Korea. New York, NY, USA: ACM, 2020, pp. 37–46, doi: 10.1145/3443279.3443304.

[21] J.-M. Xu, K.-S. Jun, X. Zhu, and A. Bellmore, "Learning from bullying traces in social media," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, Stroudsburg, PA, USA: ACL, 2012, pp. 656–666.

[22] H. Karjaluoto, P. Ulkuniemi, H. Keinanen, and O. Kuivalainen, "Antecedents of social media B2B use in industrial marketing context: Customers' view," *J. Bus. & Ind. Marketing*, 2015.

[23] S. Agrawal and A. Awekar, "Deep learning for detecting cyberbullying across multiple social media platforms," in *European Conf. Inf. Retrieval*, Cham, Switzerland: Springer International Publishing, 2018, pp. 141–153.

[24] N. Selwyn, "Social media in higher education," *Europa World of Learn.*, vol. 1, no. 3, pp. 1–10, 2012.

[25] J. F. Hair and M. Sarstedt, "Data, measurement, and causal inferences in machine learning: Opportunities and challenges for marketing," *J. Marketing Theory Pract.*, vol. 29, pp. 65–77, 2021.

[26] Z. Lv, L. Qiao, and A. K. Singh, "Advanced machine learning on cognitive computing for human behavior analysis," *IEEE Trans. Comput. Social Syst.*, vol. 8, no. 5, pp. 1194–1202, Oct. 2021.

[27] D. Ottosson, "Cyberbullying detection on social platforms using Large Language Models," Dissertation, 2023. Retrieved: https://urn.kb.se/resolve?urn=urn:nbn:se:miun:diva-48990

[28] A. Alhloul and A. Alam, "Bullying tweets detection using CNN-attention," Jan. 2023, available at SSRN 4338998.

[29] J. Wang, K. Fu, and C. T. Lu, "SOSNet: A graph convolutional network approach to fine-grained cyberbullying detection," in *Proc. IEEE Int. Conf. Big Data (Big Data),* Piscataway, NJ, USA: IEEE Press, 2020, pp. 1699–1708.

[30] H. Qudah, M. A. Alhija, and H. Tarawneh, "Improving cyberbullying detection through adaptive external dictionary in machine learning," Preprint, Sep. 2023.

[31] S. A. Mathur, S. Isarka, B. Dharmasivam, and C. D. Jaidhar, "Analysis of tweets for cyberbullying detection," in *Proc. 3rd Int. Conf. Secure Cyber Comput. Commun. (ICSCCC),* Piscataway, NJ, USA: IEEE Press, 2023, pp. 269–274.

[32] B. G. Bokolo and Q. Liu, "Cyberbullying detection on social media using machine learning," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS),* Piscataway, NJ, USA: IEEE Press, 2023, pp. 1–6.

[33] M. Nisha and J. Jebathangam, "Detection and classification of cyberbullying in social media using text mining," in *Proc. 6th Int. Conf. Electron., Commun. Aerosp. Technol.,* Piscataway, NJ, USA: IEEE Press, 2022, pp. 856–861.

[34] M. I. Mahmud, M. Mamun, and A. Abdelgawad, "A deep analysis of textual features based cyberbullying detection using machine learning," in *Proc. IEEE Global Conf. Artif. Intell. Internet Things (GCAIoT),* Piscataway, NJ, USA: IEEE Press, 2022, pp. 166–170.

[35] A. Muneer, A. Alwadain, M. G. Ragab, and A. Alqushaibi, "Cyberbullying detection on social media using stacking ensemble learning and enhanced BERT," *Information*, vol. 14, no. 8, 2023, Art. no. 467.

[36] F. Alrowais et al., "RoBERTaNET: Enhanced RoBERTa transformer based model for cyberbullying detection with GloVe features," *IEEE Access*, vol. 12, pp. 58950–58959, 2024.

[37] S. M. Fati, A. Muneer, A. Alwadain, and A. O. Balogun, "Cyberbullying detection on Twitter using deep learning-based attention mechanisms and continuous bag of words feature extraction," *Mathematics*, vol. 11, no. 16, 2023, Art. no. 3567.

[38] "Cyberbullying classification dataset." AndrewMVD. Accessed: Jan. 5, 2024. [Online]. Available: https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification

[39] S. Deegalla and H. Bostrom, "Reducing high-dimensional data by principal component analysis vs. random projection for nearest neighbor classification," in *Proc. 5th Int. Conf. Mach. Learn. Appl. (ICMLA)*, 2006, pp. 245–250.

[40] D. Hou, J. Zhang, Z. Yang, S. Liu, P. Huang, and G. Zhang, "Distribution water quality anomaly detection from UV optical sensor monitoring data by integrating principal component analysis with chi-square distribution," *Opt. Exp.*, vol. 23, no. 13, p. 17487–17510, Jun. 2015, doi: 10.1364/OE.23.017487.

[41] U. Naseem, I. Razzak, M. Khushi, P. W. Eklund, and J. Kim, "COVIDSenti: A large-scale benchmark Twitter data set for COVID-19 sentiment analysis," *IEEE Trans. Comput. Social Syst.*, vol. 8, no. 4, pp. 1003–1015, Aug. 2021.

[42] C. Suman, A. Naman, S. Saha, and P. Bhattacharyya, "A multimodal author profiling system for tweets," *IEEE Trans. Comput. Social Syst.*, vol. 8, no. 6, pp. 1407–1416, Dec. 2021.

[43] P. K. Verma, P. Agrawal, I. Amorim, and R. Prodan, "WELFake: Word embedding over linguistic features for fake news detection," *IEEE Trans. Comput. Social Syst.*, vol. 8, no. 4, pp. 881–893, Aug. 2021.

[44] S. F. Abdoh, M. A. Rizka, and F. A. Maghraby, "Cervical cancer diagnosis using random forest classifier with SMOTE and feature reduction techniques," *IEEE Access*, vol. 6, pp. 59475–59485, 2018.

[45] S. Zhong, Z. Bao, S. Gong, and K. Xia, "Person reidentification based on pose-invariant feature and B-KNN reranking," *IEEE Trans. Comput. Social Syst.*, vol. 8, no. 5, pp. 1272–1281, Oct. 2021.

[46] I. Rish et al., "An empirical study of the naive Bayes classifier," in *Proc. Int. Joint Conf. Artif. Intell. Workshop Empirical Methods Artif. Intell.*, vol. 3, no. 22, 2001, pp. 41–46.

[47] D. Rogers, A. Preece, M. Innes, and I. Spasić, "Real-time text classification of user-generated content on social media: Systematic review," *IEEE Trans. Comput. Social Syst.*, vol. 9, no. 4, pp. 1154–1166, Aug. 2021.

[48] C. Chang et al., "Multi-information preprocessing event extraction with BiLSTM-CRF attention for academic knowledge graph construction," *IEEE Trans. Comput. Social Syst.*, vol. 10, no. 5, pp. 2713–2724, Oct. 2023.

[49] Y. Gao, L. T. Yang, J. Yang, H. Wang, and Y. Zhao, "Attention U-Net based on Bi-ConvLSTM and its optimization for smart healthcare," *IEEE Trans. Comput. Social Syst.*, vol. 10, no. 4, pp. 1966–1974, Aug. 2023.

[50] H. Liu, P. Burnap, W. Alorainy, and M. L. Williams, "A fuzzy approach to text classification with two-stage training for ambiguous instances," *IEEE Trans. Comput. Social Syst.*, vol. 6, no. 2, pp. 227–240, Apr. 2019.

[51] P. Amrit, A. K. Singh, M. P. Singh, and A. K. Agrawal, "EmbedR-Net: Using CNN to embed mark with recovery through deep convolutional GAN for secure eHealth systems," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 1017–1022, Nov. 2023.

[52] A. Kumar, J. P. Singh, and A. K. Singh, "Explainable BERT-LSTM stacking for sentiment analysis of COVID-19 vaccination," *IEEE Trans. Comput. Social Syst.*, early access, doi: 10.1109/TCSS.2023.3329664.

[53] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Comput.*, vol. 10, no. 7, pp. 1895–1923, 1998.

[54] M. T. Ribeiro, S. Singh, and C. Guestrin, "'"Why should I trust you?" Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 1135–1144.

[55] A. B. Arrieta et al., "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, 2020.

[56] H. Saini, H. Mehra, R. Rani, G. Jaiswal, A. Sharma, and A. Dev, "Enhancing cyberbullying detection: A comparative study of ensemble CNN–SVM and BERT models," *Social Netw. Anal. Mining*, vol. 14, no. 1, 2023, Art. no. 1.

Open Access funding provided by 'Università degli Studi di Salerno' within the CRUI CARE Agreement